

Excel Raw Data to Insights - Complete Workflow

1. Import Raw Data

2. Data Understanding:

- Rows / Columns
- Data Types
- PK Check
- Missing Check

3. Data Cleaning:

- Remove Duplicates
- Handle Null Values
- Remove Spaces
- Fix Text
- Fix Dates
- Handle Errors
- Validate Data

4. Transformation:

- New Columns
- Split / Merge
- Categories

5. Analysis:

- Formulas
- Pivot Table
- Sorting

6. Visualization:

- KPI
- Bar Chart
- Line Chart
- Pie Chart
- Map

7. Dashboard

8. Insights & Recommendations

EXPLANATION:

1. DATA UNDERSTANDING:

- **Understand Dataset:**

- Rows
- Columns
- Column Names
- Meaning

- **Check Data Types:**

- Date
- Number
- Text

- **Identify Primary Key (PK)**

- **Check Missing Values**

- **PK Check Formula:**

=COUNTIF(A:A,A2)

- **Result:**

1 = Unique

>1 = Duplicate

2. DATA CLEANING - 7 IMPORTANT STEPS:

- **1. Remove Duplicates:**

Data → Remove Duplicates → Select PK Column

- **2. Handle Missing Values:**

Numerical → Average, Categorical → Mode

- **3. Remove Extra Spaces:**

=TRIM(A2)

- **4. Correct Text Format:**

=PROPER(A2)

=UPPER(A2)

=LOWER(A2)

- **5. Fix Date Format:**

Format Cells → Date → dd-mm-yyyy

- **6. Handle Errors:**

=ISERROR(A2)

=IFERROR(A2,0) - Fix using

- **7. Validate Data:**

=IF(A2<0,'Wrong','Correct')

3. DATA TRANSFORMATION:

- **1. Create New Columns :**

- Example Profit Margin = Profit / Sales

- **2. Split Columns:**

- Data → Text To Columns

- **3. Merge Columns:**

=A2 & B2

- **4. Create Categories:**

=IF(A2>5000,'High','Low')

4. DATA ANALYSIS:

- **1. Descriptive Analysis:**

=SUM()

=AVERAGE()

=MAX()

=MIN()

=COUNT()

- **2. Filtering and Sorting - Example Top Customers**

5.PIVOT TABLE ANALYSIS:

Insert → Pivot Table

6. DATA VISUALIZATION:

- **1. KPI Cards:**

- Sales, Profit, Orders

- **2. Column Chart:**

- Category Analysis

- **3. Line Chart:**

- Sales Trend

- **4. Pie / Donut Chart:**

- Percentage

- **5. Map Chart:**

- Location Analysis

6. DASHBOARD CREATION:

- **1. Add Charts**

- **2. Add Slicers:**

- Year, Category, Region

- **3. Dashboard Layout:**

- Top → KPI Cards
- Middle → Charts
- Bottom → Tables

7. INSIGHTS GENERATION:

- **Convert Analysis into Business Decisions**

- Example: Laptop Sales Highest → Focus on Laptop Category

- **Low Profit Region**

- Improve Marketing Strategy

FINAL WORKFLOW:

• Raw Data → Understanding → Cleaning → Transformation → Analysis → Visualization → Dashboard → Insights